Chapter 3: Optimal State Values and the Bellman Optimality Equation

Mathematical Foundations of Reinforcement Learning

Adapted from Shiyu Zhao's textbook
December 23, 2025

Learning goals

- Define *optimal policies* and *optimal state values*.
- Introduce and interpret the **Bellman optimality equation (BOE)**.
- Understand why BOE has a **unique** solution and how to **solve it iteratively**.
- Derive the form of a **greedy deterministic optimal policy**.
- See how rewards and discounting affect optimal behavior.

Notation (tabular MDP)

- States: $s \in \mathcal{S}$; actions: $a \in \mathcal{A}$; rewards: $r \in \mathcal{R}$.
- Transition model: $p(s' \mid s, a)$ and reward model: $p(r \mid s, a)$.
- Discount: $\gamma \in (0, 1)$.
- Policy: $\pi(a \mid s)$, with $\sum_a \pi(a \mid s) = 1$.
- Value functions:

$$v_\pi(s) := \mathbb{E}_\pi\Big[\sum_{t=0}^{\infty} \gamma^t r_{t+1} \,\Big|\, s_0 = s\Big],$$

$$q_\pi(s, a) := \mathbb{E}_\pi\Big[\sum_{t=0}^{\infty} \gamma^t r_{t+1} \,\Big|\, s_0 = s, a_0 = a\Big].$$

Outline

A tiny gridworld (Fig. 3.2)

The present chapter introduces the Bellman optimality equation, which is a special Bellman equation whose corresponding policy is optimal. The next chapter (Chapter 4) will introduce an important algorithm called value iteration, which is exactly the algorithm for solving the Bellman optimality equation as introduced in the present chapter.

Be prepared that this chapter is slightly mathematically intensive. However, it is worth it because many fundamental questions can be clearly answered.

3.1 Motivating example: How to improve policies?

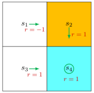


Figure 3.2: An example for demonstrating policy improvement.

Consider the policy shown in Figure 3.2. Here, the orange and blue cells represent the forbidden and target areas, respectively. The policy here is *not good* because it selects a_2 (rightward) in state s_1. How can we improve the given policy to obtain a better policy? The answer lies in state values and action values.

- *Intuition:* It is intuitively clear that the policy can improve if it selects a_3 (downward) instead of a_2 (rightward) at s_1. This is because moving downward enables the agent to avoid entering the forbidden area.
- *Mathematics:* The above intuition can be realized based on the calculation of state values and action values.

  First, we calculate the state values of the given policy. In particular, the Bellman equation of this policy is

$$\begin{aligned}
v_\pi(s_1) &= -1 + \gamma v_\pi(s_2),\\
v_\pi(s_2) &= +1 + \gamma v_\pi(s_4),\\
v_\pi(s_3) &= +1 + \gamma v_\pi(s_4),\\
v_\pi(s_4) &= +1 + \gamma v_\pi(s_4).
\end{aligned}$$

Bellman equations for the given policy

The notes write the Bellman equations for the policy in Fig. 3.2 as

$$
\begin{aligned}
v_\pi(s_1) &= -1 + \gamma v_\pi(s_2), \\
v_\pi(s_2) &= +1 + \gamma v_\pi(s_4), \\
v_\pi(s_3) &= +1 + \gamma v_\pi(s_4), \\
v_\pi(s_4) &= +1 + \gamma v_\pi(s_4).
\end{aligned}
$$

Solving the values (step-by-step)

Assume $\gamma = 0.9$.

Solve $v_\pi(s_4)$

$$v_\pi(s_4) = 1 + \gamma v_\pi(s_4) \quad \Rightarrow \quad (1-\gamma)v_\pi(s_4) = 1 \quad \Rightarrow \quad v_\pi(s_4) = \frac{1}{1-\gamma} = \frac{1}{0.1} = 10.$$

Back-substitute

$$\begin{aligned}
v_\pi(s_2) &= 1 + \gamma v_\pi(s_4) = 1 + 0.9 \cdot 10 = 10, \\
v_\pi(s_3) &= 1 + \gamma v_\pi(s_4) = 10, \\
v_\pi(s_1) &= -1 + \gamma v_\pi(s_2) = -1 + 0.9 \cdot 10 = 8.
\end{aligned}$$

Action values at s_1 and greedy improvement

Using the computed v_π values, the notes compute

$$
\begin{aligned}
q_\pi(s_1, a_1) &= -1 + \gamma v_\pi(s_1) = -1 + 0.9 \cdot 8 = 6.2, \\
q_\pi(s_1, a_2) &= -1 + \gamma v_\pi(s_2) = -1 + 0.9 \cdot 10 = 8, \\
q_\pi(s_1, a_3) &= 0 + \gamma v_\pi(s_3) = 0.9 \cdot 10 = 9, \\
q_\pi(s_1, a_4) &= -1 + \gamma v_\pi(s_1) = 6.2, \\
q_\pi(s_1, a_5) &= 0 + \gamma v_\pi(s_1) = 0.9 \cdot 8 = 7.2.
\end{aligned}
$$

Therefore a_3 is greedy: $q_\pi(s_1, a_3) \geq q_\pi(s_1, a_i)$ for all $i \neq 3$, so we update the policy to choose a_3 at s_1.

Takeaway: Greedy improvement principle

Core idea

If, at a state s, we switch the policy to choose an action with higher action-value, we can obtain a better policy (policy improvement).

- The example is simple (only s_1 is "wrong"), but motivates a general question:
- Can we characterize the best achievable value (and policies) globally?

Outline

Comparing policies by state values

Given two policies π_1, π_2:

- If $v_{\pi_1}(s) \geq v_{\pi_2}(s)$ for all $s \in \mathcal{S}$, then π_1 is *better* than π_2.
- An optimal policy is better than all other policies.

Definition: Optimal policy and optimal state value

Definition (Optimal policy and optimal state value)

A policy π^* is **optimal** if

$$v_{\pi^*}(s) \geq v_\pi(s) \quad \text{for all } s \in \mathcal{S} \text{ and for any policy } \pi.$$

The state values of π^* are called the **optimal state values**.

Fundamental questions

- **Existence:** does π^* exist?
- **Uniqueness:** is π^* unique?
- **Stochasticity:** is π^* stochastic or deterministic?
- **Algorithm:** how do we compute v^* and π^*?

Outline

Elementwise Bellman optimality equation

For every $s \in \mathcal{S}$, the notes define the BOE as

$$\begin{aligned} v(s) &= \max_{\pi(s)\in\Pi(s)} \sum_{a\in\mathcal{A}} \pi(a \mid s)\Big(\sum_{r\in\mathcal{R}} p(r \mid s, a)\, r + \gamma \sum_{s'\in\mathcal{S}} p(s' \mid s, a)\, v(s')\Big) \\ &= \max_{\pi(s)\in\Pi(s)} \sum_{a\in\mathcal{A}} \pi(a \mid s)\, q(s, a), \end{aligned}$$

where the (one-step lookahead) action-value is

$$q(s, a) := \sum_{r\in\mathcal{R}} p(r \mid s, a) r + \gamma \sum_{s'\in\mathcal{S}} p(s' \mid s, a)\, v(s').$$

Two unknowns? Resolve π first, then v

The BOE seems to involve both $v(s)$ and $\pi(a \mid s)$.

Strategy

1. For a fixed v, solve the maximization over π (a simplex-constrained optimization).
2. Plug the maximizing π back into the equation, giving a fixed-point equation in v.

Example 3.1: Maximization inside a fixed-point equation

Consider the scalar equation

$$x = \max_{y \in \mathbb{R}} \left(2x - 1 - y^2\right).$$

Step 1: maximize over y

For any fixed x, $2x - 1 - y^2$ is maximized at $y^* = 0$, giving $\max_y(2x - 1 - y^2) = 2x - 1$.

Step 2: solve for x

Substitute $y^* = 0$:

$$x = 2x - 1 \;\Rightarrow\; x = 1.$$

So the solution is $(x^*, y^*) = (1, 0)$.

Example 3.2: Maximize a convex combination

Given $q_1, q_2, q_3 \in \mathbb{R}$, choose c_1, c_2, c_3 to

$$\max \sum_{i=1}^{3} c_i q_i \quad \text{s.t.} \quad c_i \geq 0, \sum_{i=1}^{3} c_i = 1.$$

Solution

If (wlog) $q_3 \geq q_1, q_2$, then the optimal solution is $c_3^* = 1$ and $c_1^* = c_2^* = 0$.

Proof sketch

$$q_3 = (c_1 + c_2 + c_3) q_3 = c_1 q_3 + c_2 q_3 + c_3 q_3 \geq c_1 q_1 + c_2 q_2 + c_3 q_3.$$

Apply Example 3.2 to BOE: greedy maximization

Since $\sum_a \pi(a \mid s) = 1$ and $\pi(a \mid s) \geq 0$, the RHS is a convex combination:

$$\sum_a \pi(a \mid s) q(s, a).$$

Thus,

$$\sum_a \pi(a \mid s) q(s, a) \leq \sum_a \pi(a \mid s) \max_{a'} q(s, a') = \max_{a'} q(s, a').$$

A maximizing policy (statewise)

Let $a^*(s) \in \arg\max_a q(s, a)$. Then the deterministic policy

$$\pi(a \mid s) = \begin{cases} 1, & a = a^*(s), \\ 0, & \text{otherwise}, \end{cases}$$

achieves the maximum.

Matrix-vector BOE and fixed-point map

Stack values in $v \in \mathbb{R}^{|\mathcal{S}|}$. The notes give

$$v = \max_{\pi \in \Pi}(r_\pi + \gamma P_\pi v),$$

where $\max_\pi$ is taken elementwise.

- $[r_\pi]_s := \sum_a \pi(a \mid s) \sum_r p(r \mid s, a) r$.
- $[P_\pi]_{s,s'} := \sum_a \pi(a \mid s) p(s' \mid s, a)$.

Define the nonlinear map

$$f(v) := \max_{\pi \in \Pi}(r_\pi + \gamma P_\pi v).$$

Then the BOE is the fixed-point equation $v = f(v)$.

Outline

Fixed points and contractions

Fixed point

A point x^* is a fixed point of f if $f(x^*) = x^*$.

Contraction mapping

f is a contraction if there exists $\gamma \in (0, 1)$ s.t.

$$\|f(x_1) - f(x_2)\| \leq \gamma \|x_1 - x_2\| \quad \forall x_1, x_2.$$

Theorem 3.1: Contraction mapping theorem

For equations of the form $x = f(x)$, if f is a contraction mapping, then:

- **Existence:** a fixed point x^* exists.
- **Uniqueness:** the fixed point is unique.
- **Algorithm:** iterating $x_{k+1} = f(x_k)$ converges to x^* from any x_0.

Proof idea (Box 3.1): show $\{x_k\}$ is Cauchy

Let $x_{k+1} = f(x_k)$ and assume contraction factor $\gamma \in (0, 1)$.

Key inequality

For any $m > n$,

$$\begin{aligned} \|x_m - x_n\| &\leq \|x_m - x_{m-1}\| + \cdots + \|x_{n+1} - x_n\| \\ &\leq \gamma^{m-1}\|x_1 - x_0\| + \cdots + \gamma^n\|x_1 - x_0\| \\ &\leq \frac{\gamma^n}{1-\gamma}\|x_1 - x_0\|. \end{aligned}$$

Since $\gamma^n \to 0$, this shows Cauchy $\Rightarrow$ convergence.

Proof idea (Box 3.1): fixed point + uniqueness

Limit is a fixed point

If $x_k \to x^*$ and $\|f(x_k) - x_k\| = \|x_{k+1} - x_k\| \to 0$, then taking limits yields $f(x^*) = x^*$.

Uniqueness

If $f(x') = x'$ and $f(x^*) = x^*$ then

$$\|x' - x^*\| = \|f(x') - f(x^*)\| \leq \gamma \|x' - x^*\|,$$

which implies $x' = x^*$.

Outline

Theorem 3.2: $f(v)$ is a contraction

Using the max norm $\|\cdot\|_\infty$, the notes show that for all $v_1, v_2 \in \mathbb{R}^{|\mathcal{S}|}$,

$$\|f(v_1) - f(v_2)\|_\infty \leq \gamma \|v_1 - v_2\|_\infty .$$

- The contraction factor is exactly the discount rate $\gamma \in (0, 1)$.
- Consequence: BOE $v = f(v)$ has a unique fixed point.

Proof sketch (Box 3.2): compare greedy policies for v_1, v_2

Let

$$\pi_1^* := \arg\max_\pi (r_\pi + \gamma P_\pi v_1), \qquad \pi_2^* := \arg\max_\pi (r_\pi + \gamma P_\pi v_2).$$

Then elementwise,

$$\begin{aligned} f(v_1) &= r_{\pi_1^*} + \gamma P_{\pi_1^*} v_1 \geq r_{\pi_2^*} + \gamma P_{\pi_2^*} v_1, \\ f(v_2) &= r_{\pi_2^*} + \gamma P_{\pi_2^*} v_2 \geq r_{\pi_1^*} + \gamma P_{\pi_1^*} v_2. \end{aligned}$$

Subtracting and rearranging yields sandwich bounds of the form

$$\gamma P_{\pi_2^*}(v_1 - v_2) \leq f(v_1) - f(v_2) \leq \gamma P_{\pi_1^*}(v_1 - v_2).$$

Proof sketch (Box 3.2): from sandwich to ℓ_∞ contraction

Define elementwise

$$z := \max\{\, |\gamma P_{\pi_2^*}(v_1 - v_2)|,\ |\gamma P_{\pi_1^*}(v_1 - v_2)| \,\} \geq 0.$$

Then $|f(v_1) - f(v_2)| \leq z$ so $\|f(v_1) - f(v_2)\|_\infty \leq \|z\|_\infty$.
For each coordinate i, the ith row of P_π is a probability vector, so

$$|p_i^\top(v_1 - v_2)| \leq p_i^\top|v_1 - v_2| \leq \|v_1 - v_2\|_\infty .$$

Thus $\|z\|_\infty \leq \gamma\,\|v_1 - v_2\|_\infty$, proving

$$\|f(v_1) - f(v_2)\|_\infty \leq \gamma\,\|v_1 - v_2\|_\infty .$$

Outline

Theorem 3.3: Existence, uniqueness, and value iteration

Since f is a contraction, the BOE has a *unique* solution v^* and it can be computed via

$$v_{k+1} = f(v_k) = \max_{\pi \in \Pi}(r_\pi + \gamma P_\pi v_k), \qquad k = 0, 1, 2, \ldots$$

Moreover, $v_k \to v^*$ exponentially fast for any initial v_0.

Statewise value iteration update (expanded)

From the elementwise BOE, the iterative update can also be written as

$$
\begin{aligned}
v_{k+1}(s) &= \max_{\pi(s)} \sum_a \pi(a \mid s) \Big(\sum_r p(r \mid s, a) r + \gamma \sum_{s'} p(s' \mid s, a) v_k(s') \Big) \\
&= \max_{a \in \mathcal{A}} \Big(\sum_r p(r \mid s, a) r + \gamma \sum_{s'} p(s' \mid s, a) v_k(s') \Big).
\end{aligned}
$$

Interpretation

One-step lookahead: choose the action that maximizes expected immediate reward plus discounted value of next state.

Algorithm: Value iteration (pseudo-code)

```
Input: model p(s'|s,a), p(r|s,a), discount gamma in (0,1)
Initialize v(s) arbitrarily for all states s
Repeat until convergence:
  For each state s:
    v_new(s) = max_a [ sum_r p(r|s,a) r  +  gamma * sum_{s'} p(s'|s,a)
  v = v_new
Output: v ~= v*; derive greedy policy from v
```

Recovering an optimal policy from v^*

Once v^* is found, the notes define an optimal policy as

$$\pi^* = \arg\max_{\pi \in \Pi} \left(r_\pi + \gamma P_\pi v^* \right).$$

Then v^* satisfies the (standard) Bellman equation for π^*:

$$v^* = r_{\pi^*} + \gamma P_{\pi^*} v^*.$$

So the BOE is a special Bellman equation whose corresponding policy is optimal.

Theorem 3.4: Optimality of (v^*, π^*)

Statement

The unique solution v^* of the BOE is the *optimal state value*. Moreover, any π^* that achieves the maximization is an *optimal policy*. For any policy π,

$$v^* = v_{\pi^*} \geq v_\pi \quad \text{(elementwise)}.$$

Proof sketch (Box 3.3): monotone inequality chain

Start from the Bellman equation of any policy π:

$$v_\pi = r_\pi + \gamma P_\pi v_\pi.$$

From BOE optimality,

$$v^* = \max_\pi (r_\pi + \gamma P_\pi v^*) \geq r_\pi + \gamma P_\pi v^*.$$

Subtract the two equations:

$$v^* - v_\pi \geq \gamma P_\pi (v^* - v_\pi).$$

Repeat:

$$v^* - v_\pi \geq \gamma^n P_\pi^n (v^* - v_\pi) \xrightarrow[n \to \infty]{} 0$$

because $\gamma < 1$ and P_π^n is a nonnegative matrix with bounded entries. Thus $v^* \geq v_\pi$.

Outline

Theorem 3.5: Greedy deterministic optimal policy

Define the optimal one-step lookahead

$$q^*(s,a) := \sum_{r \in \mathcal{R}} p(r \mid s,a) r + \gamma \sum_{s' \in \mathcal{S}} p(s' \mid s,a)\, v^*(s').$$

Then a greedy deterministic policy is optimal:

$$\pi^*(a \mid s) = \begin{cases} 1, & a = a^*(s), \\ 0, & a \neq a^*(s), \end{cases} \qquad a^*(s) \in \arg\max_{a \in \mathcal{A}} q^*(s,a).$$

Why greedy works (proof idea)

At each state s, the BOE maximizes

$$\sum_{a \in \mathcal{A}} \pi(a \mid s)\, q^*(s, a),$$

which is a convex combination of $\{q^*(s, a)\}_a$.
By Example 3.2, the maximum is achieved by putting all probability mass on any maximizing action $a^*(s)$.

Non-uniqueness and stochasticity

- v^* is unique (fixed point of a contraction).
- π^* need not be unique: if multiple actions tie for $\max_a q^*(s,a)$, there can be many optimal policies.
- Optimal policies can be stochastic or deterministic, but there always exists a deterministic optimal policy (the greedy one).

is an optimal policy for solving the BOE. Here,

$$a^*(s) = \arg\max_a q^*(a,s),$$

where

$$q^*(s,a) \doteq \sum_{r\in\mathcal{R}} p(r|s,a)r + \gamma \sum_{s'\in\mathcal{S}} p(s'|s,a)v^*(s').$$

Box 3.4: Proof of Theorem 3.5

While the matrix-vector form of the optimal policy is $\pi^* = \arg\max_\pi(r_\pi + \gamma P_\pi v^*)$, its elementwise form is

Adapted from Shiyu Zhao's textbook

Outline

BOE shows the dependencies

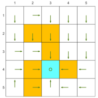
From the BOE,

$$v(s) = \max_a \Big(\underbrace{\mathbb{E}[r \mid s, a]}_{\text{reward model}} + \gamma \underbrace{\mathbb{E}[v(s') \mid s, a]}_{\text{transition model + future value}} \Big).$$

So optimal behavior is determined by

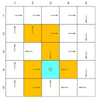
- reward values and reward model,
- discount factor γ,
- transition dynamics (the model).

Gridworld: effect of discount and reward (Fig. 3.4)



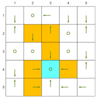
	1	2	3	4	5
1	5.8	5.6	6.2	6.5	5.8
2	6.5	7.2	8.0	7.2	6.5
3	7.2	8.0	10.0	8.0	7.2
4	8.0	10.0	10.0	10.0	8.0
5	7.2	9.0	10.0	9.0	8.1

(a) Baseline example: $r_{\text{boundary}} = r_{\text{forbidden}} = -1$, $r_{\text{target}} = 1$, $\gamma = 0.9$.



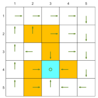
	1	2	3	4	5
1	0.0	0.0	0.0	0.0	0.0
2	0.0	0.0	0.0	0.0	0.1
3	0.0	0.0	2.0	0.1	0.1
4	0.0	2.0	2.0	2.0	0.2
5	0.0	1.0	2.0	1.0	0.5

(b) The discount rate is changed to $\gamma = 0.5$. The other parameters are the same as those in (a).



	1	2	3	4	5
1	0.0	0.0	0.0	0.0	0.0
2	0.0	0.0	0.0	0.0	0.0
3	0.0	0.0	1.0	0.0	0.0
4	0.0	1.0	1.0	1.0	0.0
5	0.0	0.0	1.0	0.0	0.0

(c) The discount rate is changed to $\gamma = 0$. The other parameters are the same as those in (a).



	1	2	3	4	5
1	3.5	3.9	4.3	4.8	5.3
2	3.1	3.5	4.8	5.3	5.9
3	2.8	2.5	10.0	5.9	6.6
4	2.5	10.0	10.0	10.0	7.3
5	2.3	9.0	10.0	9.0	8.1

(d) $r_{\text{forbidden}}$ is changed from -1 to -10. The other parameters are the same as those in (a).

Figure 3.4: The optimal policies and optimal state values given different parameter values.

Discount factor intuition

- Larger γ **(far-sighted)**: future rewards matter more; riskier short-term losses may be worth it.
- Smaller γ **(short-sighted)**: immediate reward dominates; agent tends to avoid risks.
- Extreme $\gamma = 0$: agent maximizes only immediate reward (can even fail to reach long-term goals).

Theorem 3.6: Invariance to affine reward transforms

Suppose we transform every reward by

$$r \mapsto r' = \alpha r + \beta, \qquad \alpha > 0.$$

Then the optimal state value transforms as

$$v' = \alpha v^* + \frac{\beta}{1-\gamma}\mathbf{1}.$$

Consequently, the greedy optimal policy is unchanged.

Proof sketch (Box 3.5): plug in an affine ansatz

If $r'_\pi = \alpha r_\pi + \beta \mathbf{1}$, the transformed BOE is

$$v' = \max_\pi \left(\alpha r_\pi + \beta \mathbf{1} + \gamma P_\pi v'\right).$$

Try $v' = \alpha v^* + c\mathbf{1}$. Then

$$\begin{aligned} \alpha v^* + c\mathbf{1} &= \max_\pi \left(\alpha r_\pi + \beta \mathbf{1} + \gamma P_\pi(\alpha v^* + c\mathbf{1})\right) \\ &= \max_\pi \left(\alpha r_\pi + \alpha\gamma P_\pi v^* + (\beta + c\gamma)\mathbf{1}\right), \end{aligned}$$

using $P_\pi \mathbf{1} = \mathbf{1}$. Choose $c = \beta/(1-\gamma)$ so the constant terms match.

Outline

Fig. 3.5: detour vs direct path

We next solve the new BOE in (3.9) by showing that $v' = \alpha v^* + c\mathbf{1}$ with $c = \beta/(1-\gamma)$ is a solution of (3.9). In particular, substituting $v' = \alpha v^* + c\mathbf{1}$ into (3.9) gives

$$\alpha v^* + c\mathbf{1} = \max_{\pi \in \Pi}(\alpha r_\pi + \beta\mathbf{1} + \gamma P_\pi(\alpha v^* + c\mathbf{1})) = \max_{\pi \in \Pi}(\alpha r_\pi + \beta\mathbf{1} + \alpha\gamma P_\pi v^* + c\gamma\mathbf{1}),$$

where the last equality is due to the fact that $P_\pi \mathbf{1} = \mathbf{1}$. The above equation can be reorganized as

$$\alpha v^* = \max_{\pi \in \Pi}(\alpha r_\pi + \alpha\gamma P_\pi v^*) + \beta\mathbf{1} + c\gamma\mathbf{1} - c\mathbf{1},$$

which is equivalent to

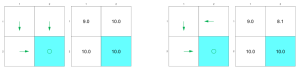
$$\beta\mathbf{1} + c\gamma\mathbf{1} - c\mathbf{1} = 0.$$

Since $c = \beta/(1-\gamma)$, the above equation is valid and hence $v' = \alpha v^* + c\mathbf{1}$ is the solution of (3.9). Since (3.9) is the BOE, v' is also the unique solution. Finally, since v' is an affine transformation of v^*, the relative relationships between the action values remain the same. Hence, the greedy optimal policy derived from v' is the same as that from v^*: $\arg\max_{\pi \in \Pi}(r_\pi + \gamma P_\pi v')$ is the same as $\arg\max_\pi(r_\pi + \gamma P_\pi v^*)$.

Readers may refer to [9] for a further discussion on the conditions under which modifications to the reward values preserve the optimal policy.

Avoiding meaningless detours

In the reward setting, the agent receives a reward of $r_{\text{other}} = 0$ for every movement step (unless it enters a forbidden area or the target area or attempts to go beyond the boundary). Since a zero reward is not a punishment, would the optimal policy take meaningless detours before reaching the target? Should we set r_{other} to be negative to encourage the agent to reach the target as quickly as possible?



(a) Optimal policy (b) Non-optimal policy

Figure 3.5: Examples illustrating that optimal policies do not take meaningless detours due to the discount rate.

Consider the examples in Figure 3.5, where the bottom-right cell is the target area

Geometric-series derivation of the returns

Assume that reaching the target yields a reward 1 each time step thereafter.

Direct arrival (no detour)

$$G = 1 + \gamma \cdot 1 + \gamma^2 \cdot 1 + \cdots = \sum_{t=0}^{\infty} \gamma^t = \frac{1}{1-\gamma}.$$

With $\gamma = 0.9$, this is 10.

Two-step detour before reaching target

$$G = 0 + \gamma \cdot 0 + \gamma^2 \cdot 1 + \gamma^3 \cdot 1 + \cdots = \gamma^2 \sum_{t=0}^{\infty} \gamma^t = \frac{\gamma^2}{1-\gamma}.$$

With $\gamma = 0.9$, this is 8.1.

Conclusion

Even if intermediate rewards are zero, discounting penalizes longer trajectories.

Outline

Summary: the chain of ideas

1. Define optimality via state values: π^* maximizes $v_\pi(s)$ for all s.
2. BOE: $v = \max_\pi(r_\pi + \gamma P_\pi v)$ and its elementwise form.
3. The max over policies yields greedy (deterministic) choices at each state.
4. $f(v)$ is a contraction $\Rightarrow$ unique fixed point v^*.
5. Value iteration ($v_{k+1} = f(v_k)$) converges to v^*.
6. Greedy w.r.t. v^* produces an optimal policy; optimal policies may not be unique.
7. Reward scaling/shifting (affine transforms) does not change optimal policies.

Checklist of included examples/derivations

- Motivating policy-improvement example: solving Bellman equations and action values.
- Example 3.1: nested maximization in a fixed-point equation.
- Example 3.2: maximizing a convex combination (why greedy solutions appear).
- Contraction mapping theorem + proof highlights (Cauchy sequence bound).
- Theorem 3.2: proof sketch that BOE map f is a contraction in ℓ_∞.
- Theorem 3.3: value iteration algorithm and convergence claim.
- Theorem 3.4: proof sketch of $v^* \geq v_\pi$ via iterated inequality.
- Theorem 3.5: greedy deterministic optimal policy and q^* definition.
- Theorem 3.6: reward affine invariance + substitution proof sketch.
- Detour example: geometric series returns $\frac{1}{1-\gamma}$ vs $\frac{\gamma^2}{1-\gamma}$.

Questions?

Thanks!